# REZON fall detector · wiring list

SM-24 geophone · ADC 15 Click · ESP32-S3 · PCM5102A + PAM8403 · SIM868 (Waveshare) · 18650 · hackathon build · 2026-11-12 · same numbering as the 3D guide (Rezon_Wiring_Guide.html)

### HOW TO READ A HOLE

- Main board: **rows A … Z, AA** (27) from the back edge to the front edge; **columns 1 … 35** from left to right, **seen from the component side** with the rails and H1 at the front.
- **K6** = row K, column 6. Lid board: rows A–H, columns 1–12, same rule.
- On the **solder side** (board turned over like a page) row A stays at the top and **column 1 is on the right**. All maps of solder-side work are drawn that way.
- Write “A” and “1” on both sides of the board before you start.

### HOW EVERY CONNECTION IS MADE

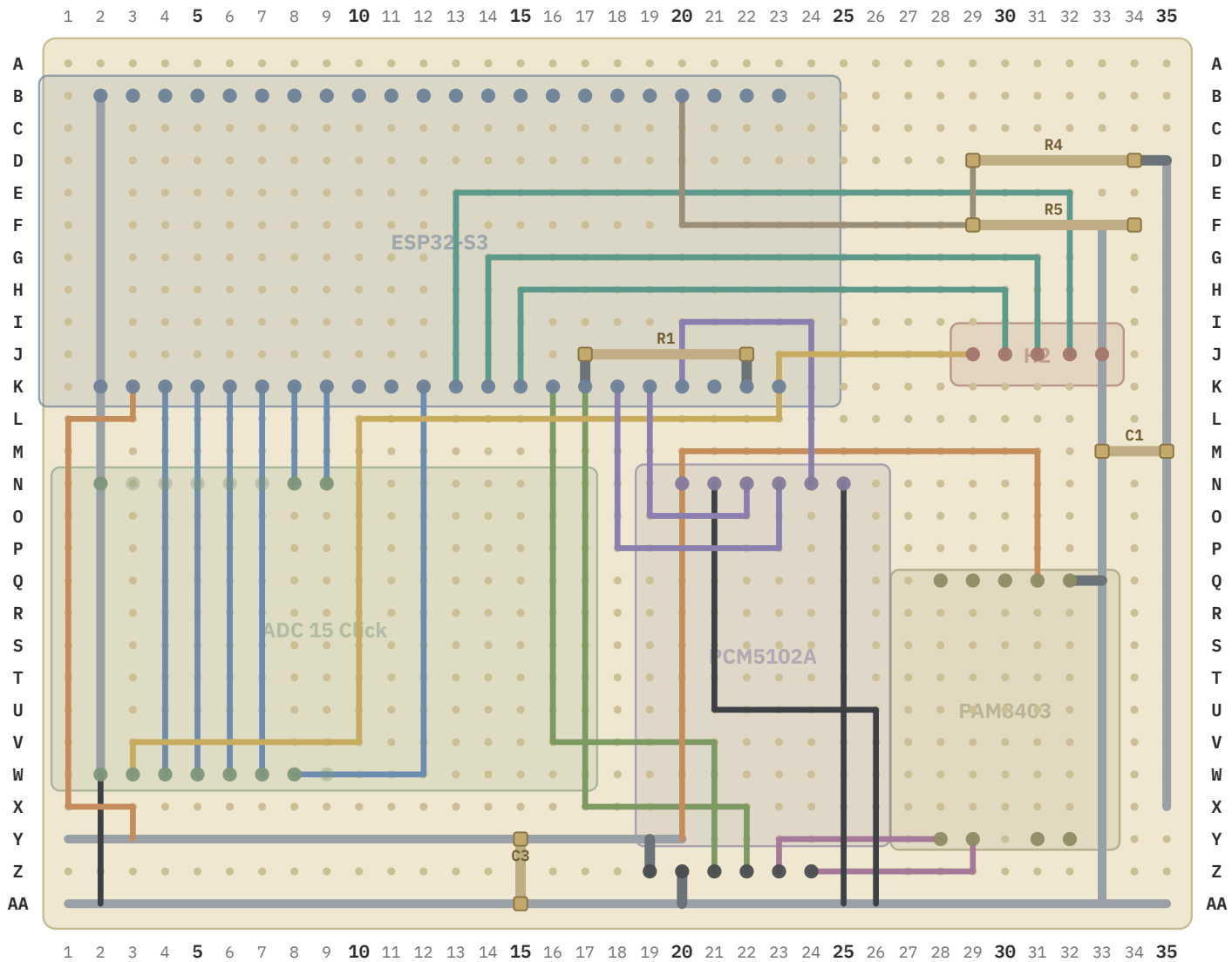
- **Bare wire** (rails/buses): 20 AWG solid, laid across the pads on the solder side, soldered at each listed hole.
- **Bridge**: bend the component leg flat onto the neighbouring pin and solder both (one joint).
- **Link** (33×): one insulated core from the 6×0.22 cable on the solder side. Cut to the listed length, strip 3 mm both ends, solder each end onto the named pin leg. “via” lists the corner holes the wire passes over.
- **Top wire**: on the component side (screw terminals, module pads).
- Two wires may cross (insulated); a link never runs along a bare rail.

### COLOURS USED IN THIS LIST

GND · 5 V · VBAT · 3V3 · SPI · I²S · UART · Speaker · LED/button · Battery sense · Bare wire

Colours are for reading the list. The insulated cores can be any colour; label both ends with tape if they are all the same.

### BOARD LAYOUT · COMPONENT SIDE




COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

**Check your modules first.** The ESP32-S3 rows must be 9 holes apart (B and K), the Click rows 9 apart (N and W). If one of yours differs, shift that module's second row and every hole that refers to it. Firmware pin map: SCK 12, MISO 13, MOSI 14, ADC-CS 11, clock-CS 9, DRDY 10, ADC-RST 8, I²S BCK 4 / LCK 5 / DIN 6, SIM868 TXD→GPIO18, GPIO17→SIM868 RXD, SIM868 PWRKEY 16 (HIGH 1.5 s), button 7, LED 15, VBAT/2 on GPIO1.

# REF Every pin hole on the main board

Use this to check a hole before soldering. Pins are listed per module, left to right on the component side.

### ESP32-S3

| | |
|---|---|
| B2 | J3-22 GND |
| B3 | J3-21 GND |
| B4 | J3-20 GPIO19 |
| B5 | J3-19 GPIO20 |
| B6 | J3-18 GPIO21 |
| B7 | J3-17 GPIO47 |
| B8 | J3-16 GPIO48 |
| B9 | J3-15 GPIO45 |
| B10 | J3-14 GPIO0 |
| B11 | J3-13 GPIO35 |
| B12 | J3-12 GPIO36 |
| B13 | J3-11 GPIO37 |
| B14 | J3-10 GPIO38 |
| B15 | J3-9 GPIO39 |
| B16 | J3-8 GPIO40 |
| B17 | J3-7 GPIO41 |
| B18 | J3-6 GPIO42 |
| B19 | J3-5 GPIO2 |
| B20 | J3-4 GPIO1 |
| B21 | J3-3 RX |
| B22 | J3-2 TX |
| B23 | J3-1 GND |
| K2 | J1-22 GND |
| K3 | J1-21 5V |
| K4 | J1-20 GPIO14 |
| K5 | J1-19 GPIO13 |
| K6 | J1-18 GPIO12 |
| K7 | J1-17 GPIO11 |
| K8 | J1-16 GPIO10 |
| K9 | J1-15 GPIO9 |
| K10 | J1-14 GPIO46 |
| K11 | J1-13 GPIO3 |
| K12 | J1-12 GPIO8 |
| K13 | J1-11 GPIO18 |
| K14 | J1-10 GPIO17 |
| K15 | J1-9 GPIO16 |
| K16 | J1-8 GPIO15 |
| K17 | J1-7 GPIO7 |
| K18 | J1-6 GPIO6 |
| K19 | J1-5 GPIO5 |
| K20 | J1-4 GPIO4 |
| K21 | J1-3 EN |
| K22 | J1-2 3V3 |
| K23 | J1-1 3V3 |

### CLICK

| | |
|---|---|
| N2 | GND |
| N3 | NC |
| N4 | NC |
| N5 | NC |
| N6 | NC |
| N7 | NC |
| N8 | INT (DRDY) |
| N9 | CS2 |
| W2 | GND |
| W3 | 3V3 |
| W4 | SDI (MOSI) |
| W5 | SDO (MISO) |
| W6 | SCK |
| W7 | CS1 |
| W8 | RST |
| W9 | NC |

### PCM5102A

| | |
|---|---|
| N20 | VIN |
| N21 | GND |
| N22 | LCK |
| N23 | DIN |
| N24 | BCK |
| N25 | SCK |

### H2

| | |
|---|---|
| J29 | 3V3 → SIM868 pin 36 |
| J30 | PWRKEY → SIM868 pin 19 |
| J31 | SIM RXD → SIM868 pin 1 |
| J32 | SIM TXD → SIM868 pin 2 |
| J33 | GND → SIM868 pin 3 |

### PAM8403

| | |
|---|---|
| Q28 | L in |
| Q29 | ⏚ in |
| Q30 | R in |
| Q31 | +5V |
| Q32 | GND |
| Y28 | L+ out |
| Y29 | L– out |
| Y31 | R– out |
| Y32 | R+ out |

### H1

| | |
|---|---|
| Z19 | 5V |
| Z20 | GND |
| Z21 | LED |
| Z22 | BTN |
| Z23 | SPK+ |
| Z24 | SPK– |

### PARTS ON THE MAIN BOARD

| | |
|---|---|
| C3 | 470 µF · Y15 +, AA15 – · step 4 |
| C1 | 1000 µF · M35 +, M33 – · step 8 |
| R1 | 10 kΩ · J17, J22 · step 9 |
| R4 | 100 kΩ · D34, D29 · step 9 |
| R5 | 100 kΩ · F29, F34 · step 9 |

### BARE RAILS AND BUSES

| | |
|---|---|
| AA1–AA35 | GND rail |
| Y1–Y20 | 5 V rail |
| D35–X35 | VBAT rail (switched battery) |
| B2–W2 | GND bus col 2 |
| F33–AA33 | GND bus col 33 |

### OFF-BOARD WIRES LAND IN

| | |
|---|---|
| X35 | VBAT from rocker pin 2 |
| W35 | VBAT to boost IN+ |
| AA35 | GND from holder – lead |
| AA34 | GND to boost IN– |
| Y1 | 5 V from boost OUT+ |
| AA1 | GND from boost OUT– |
| AA18 AA19 | GND wires up to the Click screw terminals |
| L35 | VBAT to the SIM868 battery plug (red) |
| L33 | GND to the SIM868 battery plug (black) |

## STEP 2 Cut the perfboard

Cutting bench

Cut the 9 × 15 cm perfboard (stock row 101) into the main board (35 columns × 27 rows of holes, ≈ 90 × 70 mm) and the lid board (12 × 8 holes, ≈ 30 × 20 mm). Score both sides along a hole row, snap, sand flat, knock 1 mm off each corner. With a pen, write “A” at the back edge and “1” at the left edge on BOTH sides of the main board.

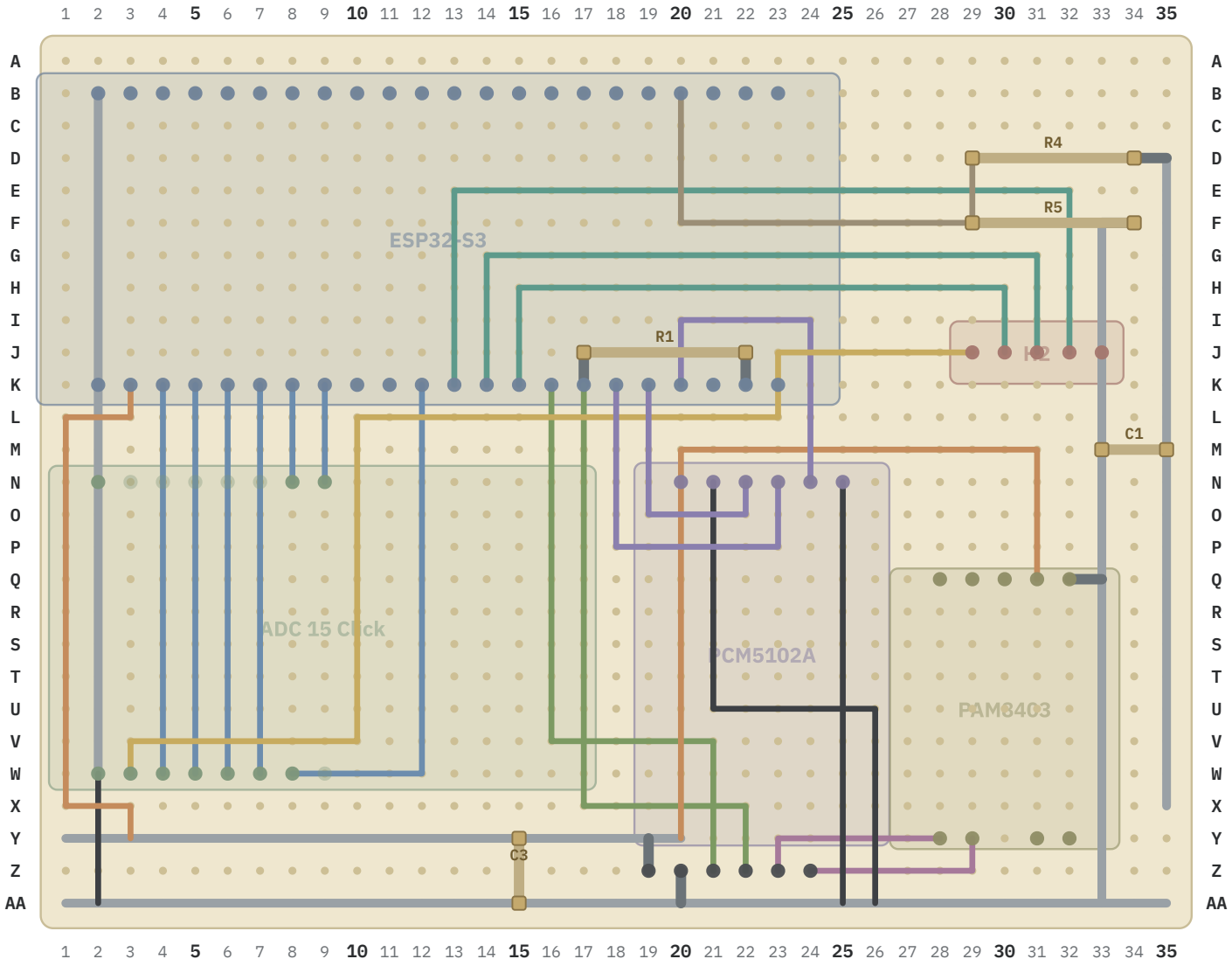
## STEP 3 Sockets, headers & module pins

Soldering station · Cutting bench

Component side up. Solder every socket and header before any wire. Plug the module in while soldering so the rows stay parallel; tack one pin at each end first, check, then solder the rest.

HOLES

| | |
|---|---|
| B2 … B23 | ESP32-S3 J3 socket, 22 contacts. B23 = J3-1 (GND, antenna end), B2 = J3-22 (GND, USB end) |
| K2 … K23 | ESP32-S3 J1 socket, 22 contacts. K23 = J1-1 (3V3), K3 = J1-21 (5V), K2 = J1-22 (GND) |
| N2 … N9 | Click back header socket, 8 contacts. N9 = CS2, N8 = INT, N7–N3 = NC, N2 = GND |
| W2 … W9 | Click front header socket, 8 contacts. W9 = NC, W8 = RST, W7 = CS1, W6 = SCK, W5 = SDO, W4 = SDI, W3 = 3V3, W2 = GND |
| J29 … J33 | H2 SIM868 header, 5 male pins pointing up: J29 3V3, J30 PWRKEY, J31 SIM RXD, J32 SIM TXD, J33 GND (the SIM868 board itself is off the perfboard, at the back wall) |
| N20 … N25 | PCM5102A: its own male header soldered straight into the board. N20 VIN, N21 GND, N22 LCK, N23 DIN, N24 BCK, N25 SCK |
| Q28 … Q32 | PAM8403 input/power pins (male header pins, module 2 mm above the board): Q28 L in, Q29 ⏚ in, Q30 R in, Q31 +5V, Q32 GND |
| Y28 Y29 Y31 Y32 | PAM8403 output pins: Y28 L+, Y29 L–, Y31 R–, Y32 R+ |
| Z19 … Z24 | H1 lid header, 6 male pins pointing up: Z19 5V, Z20 GND, Z21 LED, Z22 BTN, Z23 SPK+, Z24 SPK– |




COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

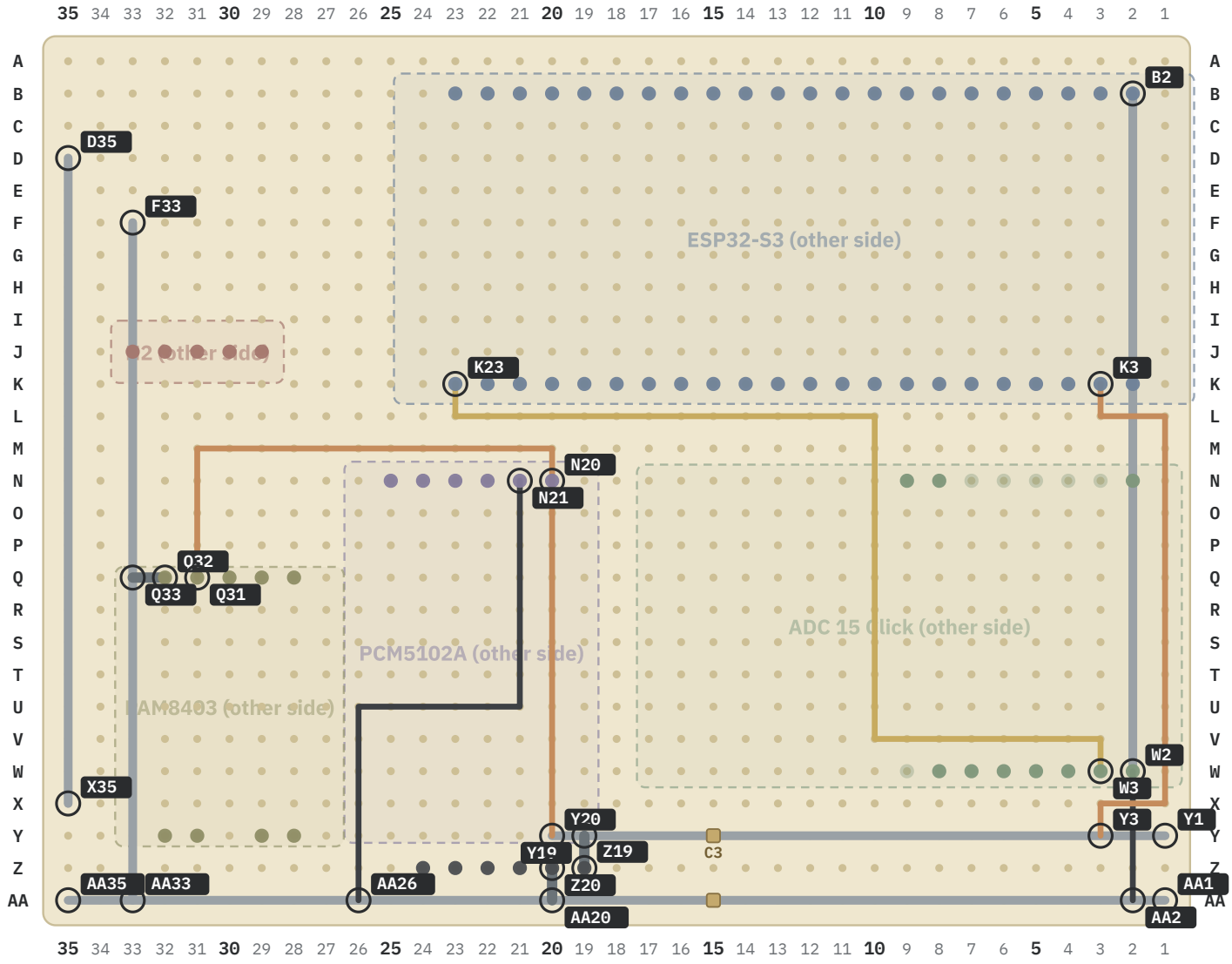
## STEP 4 Rails, buses & power distribution

Soldering station · Cutting bench

Board turned over (solder side). Lay the bare rails first, then fit C3, then the bridges and links. Bare wire: straighten 20 AWG, lay it across the pads, solder at every hole that has a pin or leg and at every 4th hole in between.

DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 4.1 | Part | **C3** 470 µF · electrolytic, stripe = − legs into **Y15** + **AA15** − (component side) | | | bend legs flat after soldering |
| 4.2 | Bare wire | **AA1** **GND rail** | **AA35** **GND rail** GND rail (bare wire along row AA) | along AA1 → AA35 solder at: AA15, AA1, AA35 | bare 20 AWG · cut 100 mm |
| 4.3 | Bare wire | **Y1** **5 V rail** | **Y20** **5 V rail** 5 V rail (bare wire along row Y) | along Y1 → Y20 solder at: Y15, Y1, Y20 | bare 20 AWG · cut 60 mm |
| 4.4 | Bare wire | **B2** **ESP32-S3** J3-22 GND | **W2** **Click** GND GND bus col 2: ESP32 GND ↔ Click GND (bare) | along B2 → W2 solder at: B2, K2, N2, W2, B2, W2 | bare 20 AWG · cut 65 mm |
| 4.5 | Bare wire | **F33** **GND bus** col 33 | **AA33** **GND rail** GND bus col 33: H2 GND, C1−, SIM868 GND lead, PAM GND → rail (bare) | along F33 → AA33 solder at: J33, M33, F33, AA33 | bare 20 AWG · cut 65 mm |
| 4.6 | Bare wire | **D35** **VBAT rail** | **X35** **VBAT rail** VBAT rail (bare wire along col 35) | along D35 → X35 solder at: M35, D35, X35 | bare 20 AWG · cut 65 mm |
| 4.7 | Bridge | **Z19** **H1** 5V | **Y19** **5 V rail** H1 5V → 5 V rail | | solder bridge / bent leg |
| 4.8 | Bridge | **Z20** **H1** GND | **AA20** **GND rail** H1 GND → GND rail | | solder bridge / bent leg |
| 4.9 | Bridge | **Q32** **PAM8403** GND | **Q33** **GND bus** col 33 PAM8403 GND → GND bus | | solder bridge / bent leg |
| 4.10 | Link | **W2** **Click** GND | **AA2** **GND rail** GND bus col 2 → GND rail | straight | insulated core · cut 20 mm, strip 3 mm both ends |
| 4.11 | Link | **Y3** **5 V rail** | **K3** **ESP32-S3** J1-21 5V 5 V rail → ESP32 5V | via X3 → X1 → L1 → L3 | insulated core · cut 55 mm, strip 3 mm both ends |
| 4.12 | Link | **Y20** **5 V rail** | **N20** **PCM5102A** VIN 5 V rail → PCM VIN | straight | insulated core · cut 40 mm, strip 3 mm both ends |
| 4.13 | Link | **N20** **PCM5102A** VIN | **Q31** **PAM8403** +5V PCM VIN → PAM8403 +5V | via M20 → M31 | insulated core · cut 50 mm, strip 3 mm both ends |
| 4.14 | Link | **N21** **PCM5102A** GND | **AA26** **GND rail** PCM GND → GND rail | via U21 → U26 | insulated core · cut 55 mm, strip 3 mm both ends |
| 4.15 | Link | **K23** **ESP32-S3** J1-1 3V3 | **W3** **Click** 3V3 ESP32 3V3 → Click 3V3 | via L23 → L10 → V10 → V3 | insulated core · cut 90 mm, strip 3 mm both ends |




SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT

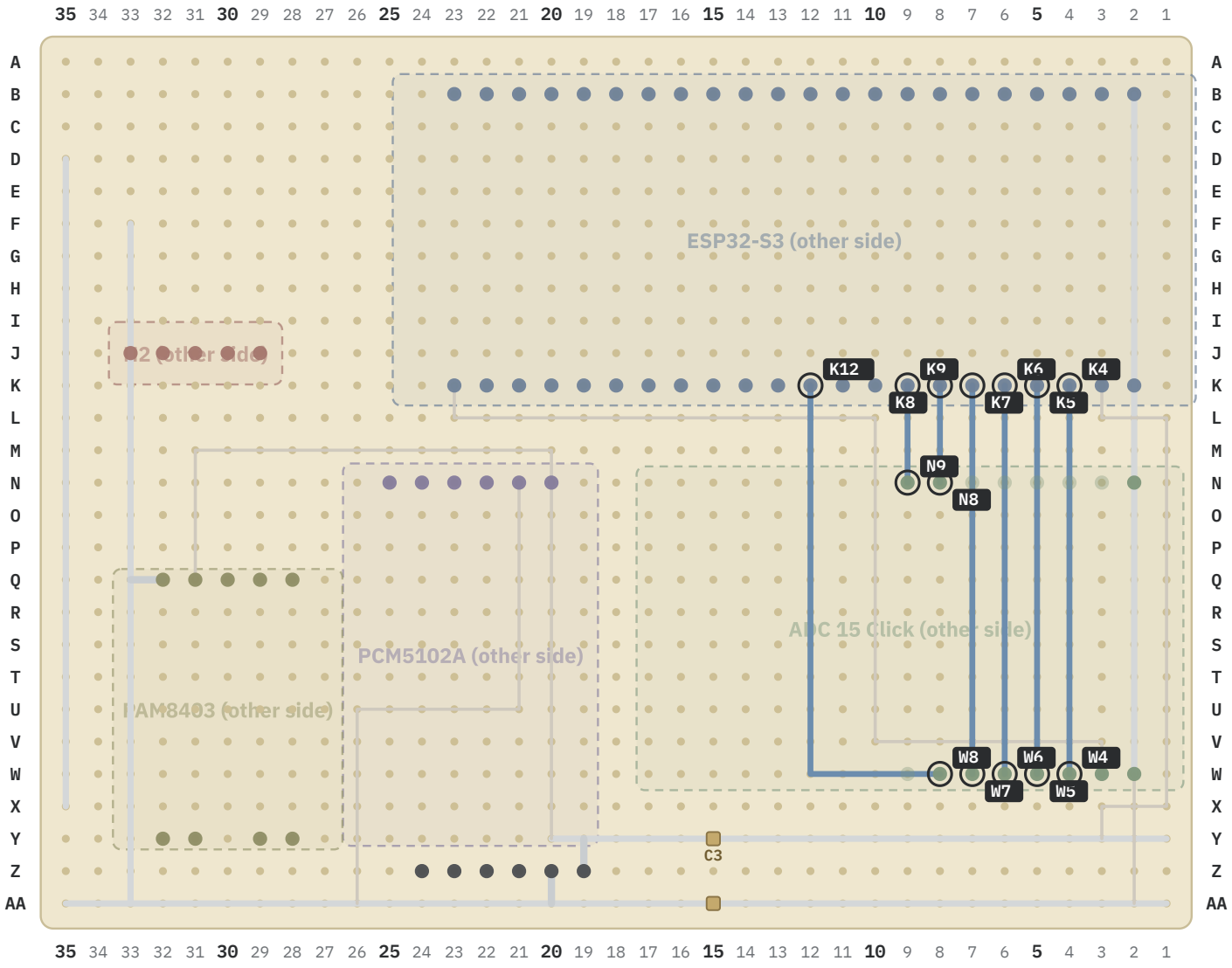
## STEP 5 ADC 15 Click ↔ ESP32-S3 (SPI)

Soldering station

Seven links on the solder side. Six run straight in one column and pass the Click's unused NC pins (N3–N7, W9): keep the insulation intact there.

DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 5.1 | Link | K6 **ESP32-S3** J1-18 GPIO12 | W6 **Click** SCK<br>SCK | straight | insulated core · cut 40 mm, strip 3 mm both ends |
| 5.2 | Link | K5 **ESP32-S3** J1-19 GPIO13 | W5 **Click** SDO (MISO)<br>MISO | straight | insulated core · cut 40 mm, strip 3 mm both ends |
| 5.3 | Link | K4 **ESP32-S3** J1-20 GPIO14 | W4 **Click** SDI (MOSI)<br>MOSI | straight | insulated core · cut 40 mm, strip 3 mm both ends |
| 5.4 | Link | K7 **ESP32-S3** J1-17 GPIO11 | W7 **Click** CS1<br>ADC chip-select | straight | insulated core · cut 40 mm, strip 3 mm both ends |
| 5.5 | Link | K9 **ESP32-S3** J1-15 GPIO9 | N9 **Click** CS2<br>clock-chip select | straight | insulated core · cut 20 mm, strip 3 mm both ends |
| 5.6 | Link | K8 **ESP32-S3** J1-16 GPIO10 | N8 **Click** INT (DRDY)<br>DRDY, active low | straight | insulated core · cut 20 mm, strip 3 mm both ends |
| 5.7 | Link | K12 **ESP32-S3** J1-12 GPIO8 | W8 **Click** RST<br>ADC reset | via W12 | insulated core · cut 50 mm, strip 3 mm both ends |




SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT

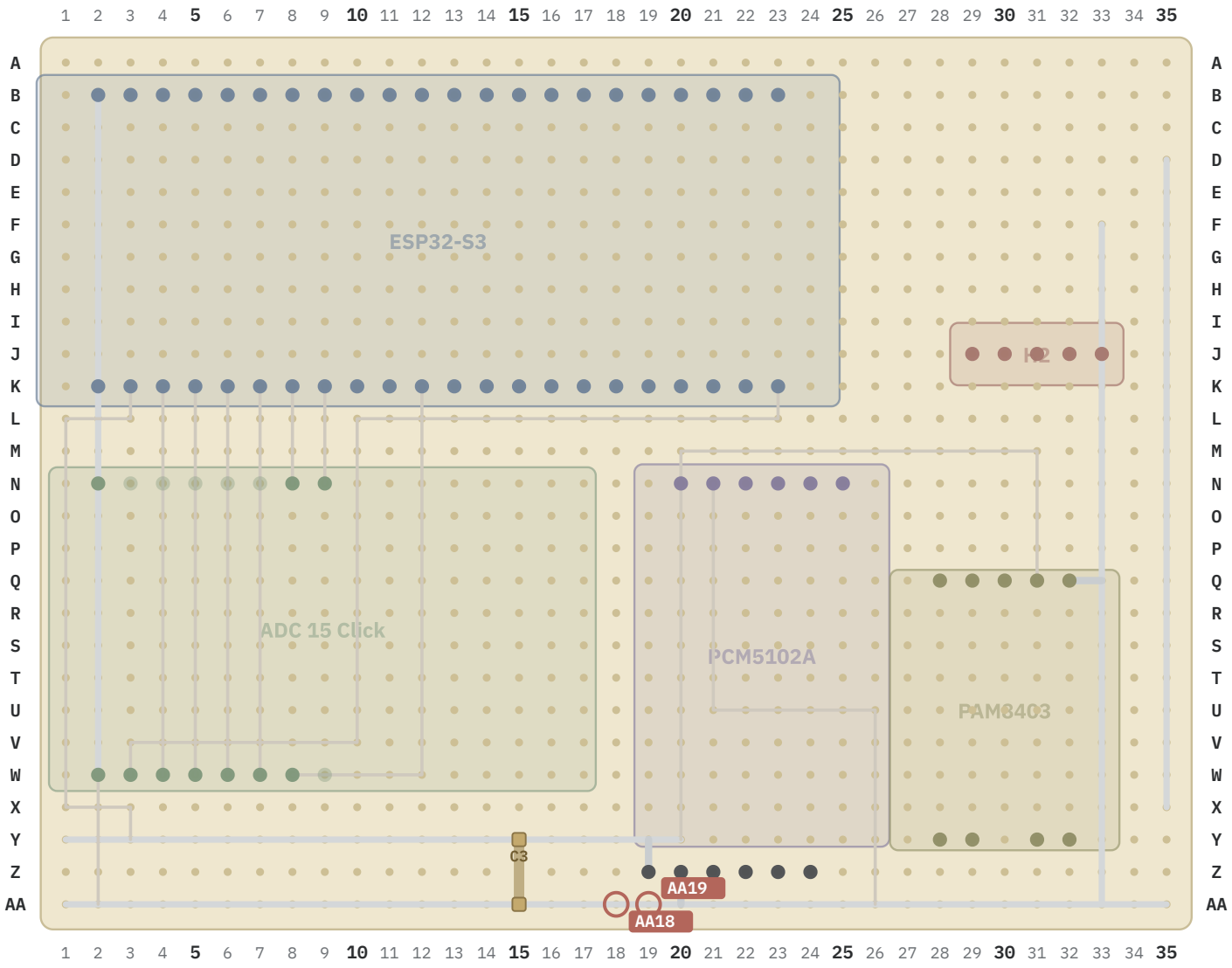
## STEP 6 Geophone → Click CH1

Soldering station | Small flat screwdriver

Solder the 1 kΩ resistor across the two SM-24 solder tabs. Twist two cores of the 6×0.22 cable (about 1 turn per cm), solder to the tabs (mark the + core). Screw-terminal ends: strip 5 mm, twist, tin lightly. Each terminal takes at most two conductors.

DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 6.1 | Pair | ↗ SM-24 + tab | ↗ Click CH1+ screw terminal | | 6×0.22 cable core · cut 115 mm, twist the two cores ~1 turn/cm |
| 6.2 | Pair | ↗ SM-24 – tab | ↗ Click CH1– screw terminal | | 6×0.22 cable core · cut 105 mm, twist the two cores ~1 turn/cm |
| 6.3 | Top wire | AA18 GND rail | ↗ Click CH1– screw terminal<br>2nd wire in CH1–: ties it to ground | | insulated core · cut 55 mm, strip 5 mm (screw end) / 3 mm |
| 6.4 | Top wire | AA19 GND rail | ↗ Click CH2– screw terminal<br>grounds the unused channel | | insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm |
| 6.5 | Top wire | ↗ Click CH2+ screw terminal | ↗ Click CH2– screw terminal<br>short loop: CH2 shorted | | insulated core · cut 20 mm, strip 5 mm (screw end) / 3 mm |




COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

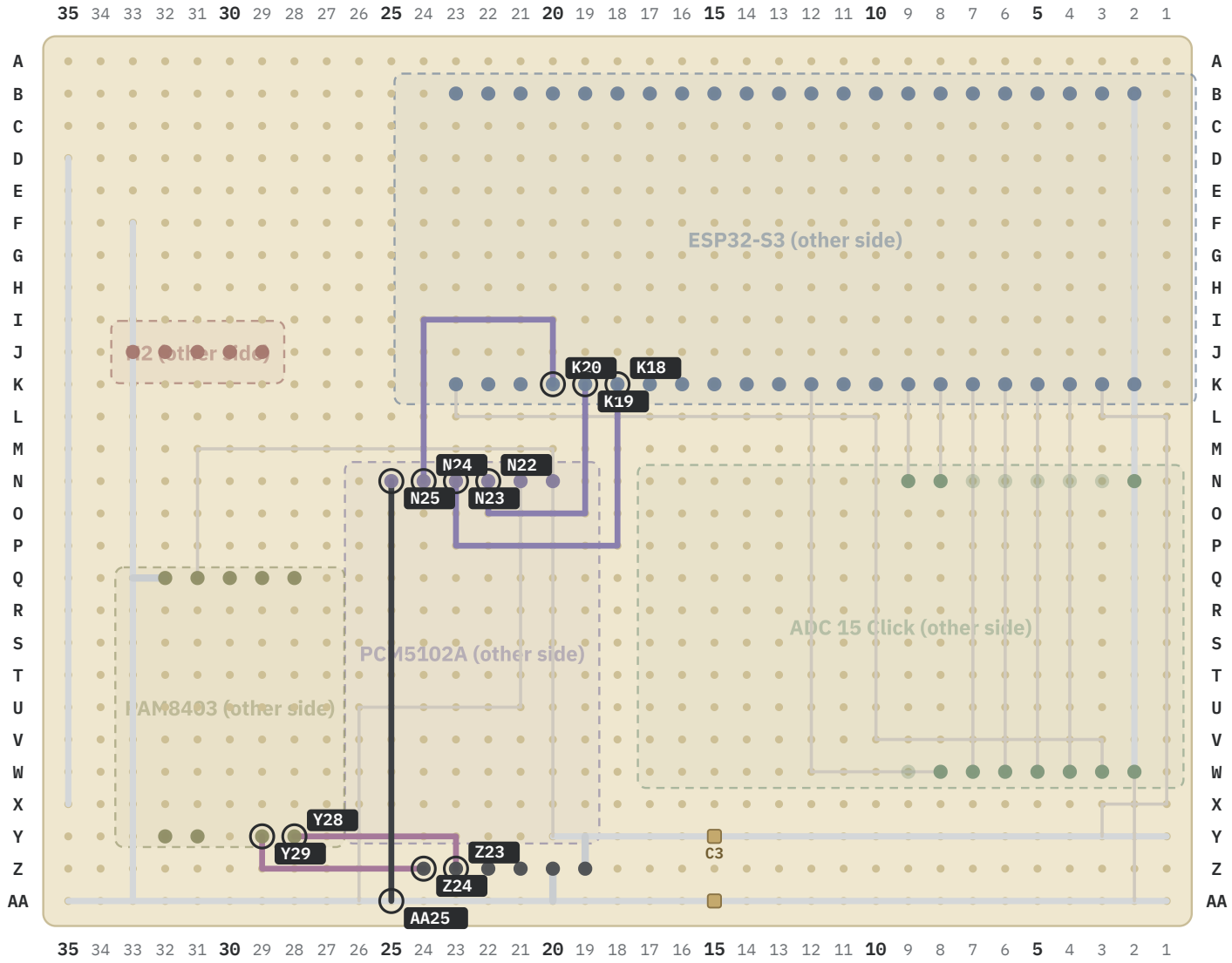
## STEP 7 Audio: I²S, DAC → amp → H1

Soldering station

Before wiring, set the PCM5102A back-side jumpers: 1 FLT = L, 2 DEMP = L, 3 XSMT = H, 4 FMT = L. The two DAC → amp wires run on the component side, pad to pad on top of the modules.

DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 7.1 | Link | Y28 **PAM8403** L+ out | Z23 **H1** SPK+ speaker + | via Y23 | insulated core · cut 25 mm, strip 3 mm both ends |
| 7.2 | Link | Y29 **PAM8403** L– out | Z24 **H1** SPK– speaker –, NOT ground | via Z29 | insulated core · cut 25 mm, strip 3 mm both ends |
| 7.3 | Link | K20 **ESP32-S3** J1-4 GPIO4 | N24 **PCM5102A** BCK bit clock | via I20 → I24 | insulated core · cut 40 mm, strip 3 mm both ends |
| 7.4 | Link | K19 **ESP32-S3** J1-5 GPIO5 | N22 **PCM5102A** LCK word clock | via O19 → O22 | insulated core · cut 30 mm, strip 3 mm both ends |
| 7.5 | Link | K18 **ESP32-S3** J1-6 GPIO6 | N23 **PCM5102A** DIN data | via P18 → P23 | insulated core · cut 40 mm, strip 3 mm both ends |
| 7.6 | Link | N25 **PCM5102A** SCK | AA25 **GND rail** SCK → GND rail (SCK must be grounded) | straight | insulated core · cut 45 mm, strip 3 mm both ends |
| 7.7 | Top wire | ↗ PCM5102A pad L (top) | ↗ PAM8403 pad L in (top, above Q28) | | insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm |
| 7.8 | Top wire | ↗ PCM5102A pad AGND (top) | ↗ PAM8403 pad ⏚ in (top, above Q29) | | insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm |




SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT

# STEP 8 SIM868: power leads, UART, jumpers

Soldering station | Cutting bench

The Waveshare Pico-SIM868 board is used as a modem only: pull the Pico H off it, it is not used. Fit C1, then the four links on the solder side to H2. Then the two 0.5 mm² power leads from L35/L33 (top side) to a PH2.0 plug for the board's battery socket, and five M-F jumpers from H2 into the board's Pico sockets. Before power-on: GSM antenna on the "GSM" u.FL socket, nano SIM (PIN off) in the slot, the board's power slide switch on the battery side.

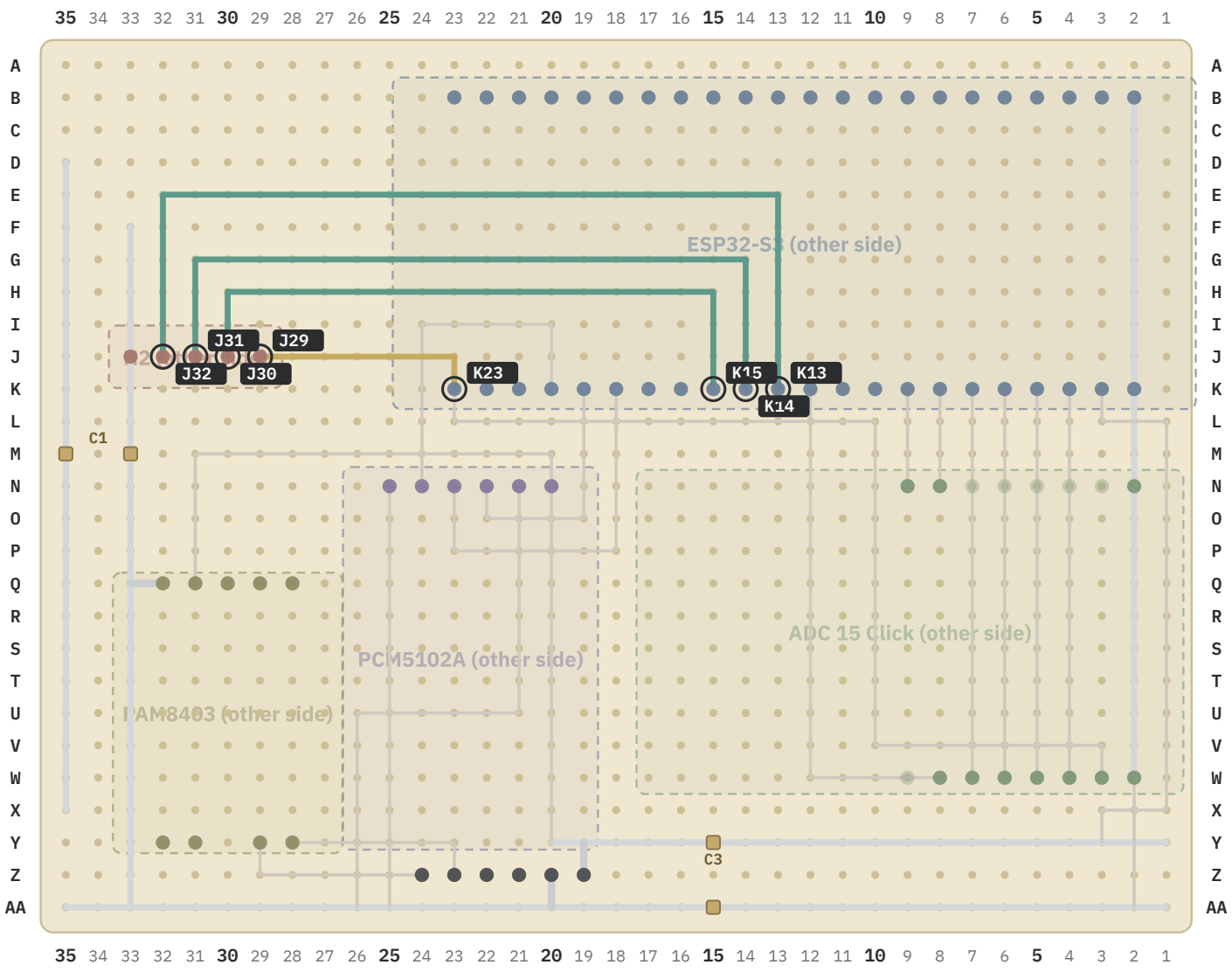
## SIM868 BOARD: WHAT GOES WHERE

| | |
|---|---|
| pin 1 (GP0) | SIM868 RXD, through the board's own 3.3 V ↔ 2.8 V level shifter. From H2 J31 ← GPIO17 (ESP TX). No divider needed. |
| pin 2 (GP1) | SIM868 TXD. To H2 J32 → GPIO18 (ESP RX). |
| pin 3 | GND. From H2 J33 (GND bus col 33). |
| pin 19 (GP14) | PWR_EN: drives an NPN on the SIM868 PWRKEY. HIGH for 1.5 s switches the module on (or off). From H2 J30 ← GPIO16. |
| pin 36 (3V3 OUT) | Powers the board's level shifter, microphone and RTC backup. From H2 J29 ← ESP32 3V3. Without it the UART stays dead. |
| pin 39, pin 40 | VSYS, VBUS: leave EMPTY. Never 5 V into this board. |
| BAT (PH2.0) | Battery socket → SIM868 VBAT (3.4–4.4 V, 2 A bursts). Red lead from L35 (VBAT rail), black from L33 (GND bus). Check polarity against the plug of the Li-po in the box. |
| GSM (u.FL) | GSM antenna, stuck flat to the inner wall at the back left. GNSS and BT sockets stay empty. |

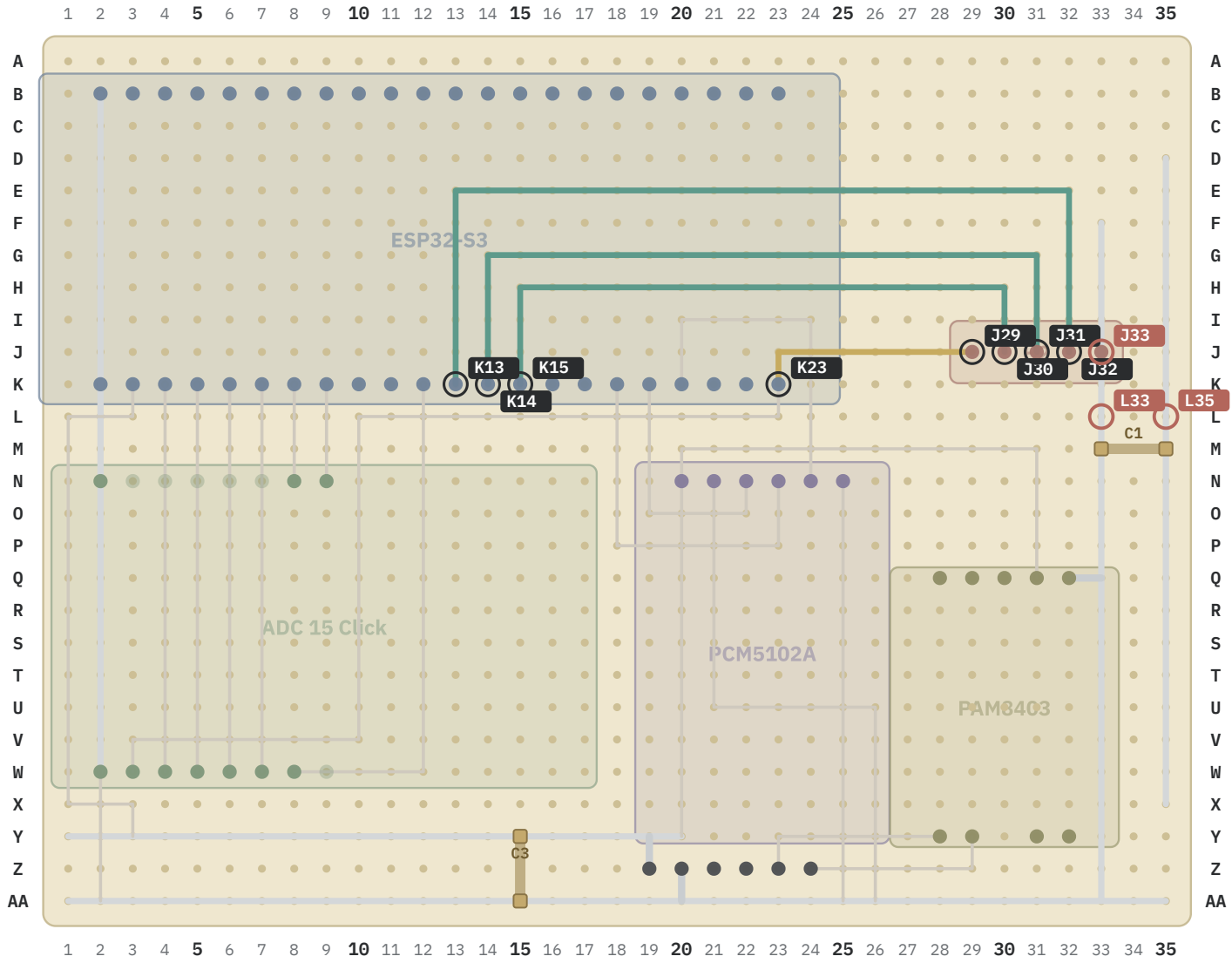
## DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 8.1 | Part | **C1** 1000 µF · electrolytic, stripe = –, burst reserve at the SIM868 power take-off<br>legs into M35 + M33 – (component side) | | | bend legs flat after soldering |
| 8.2 | Link | K23 **ESP32-S3** J1-1 3V3 | J29 **H2** 3V3 → SIM868 pin 36<br>ESP32 3V3 (J1-1) → H2 3V3 (feeds the SIM868 level shifter) | via J23 | insulated core · cut 30 mm, strip 3 mm both ends |
| 8.3 | Link | K15 **ESP32-S3** J1-9 GPIO16 | J30 **H2** PWRKEY → SIM868 pin 19<br>GPIO16 → H2 PWRKEY (HIGH 1.5 s = power on/off) | via H15 → H30 | insulated core · cut 60 mm, strip 3 mm both ends |
| 8.4 | Link | K14 **ESP32-S3** J1-10 GPIO17 | J31 **H2** SIM RXD → SIM868 pin 1<br>GPIO17 (ESP TX) → H2 SIM RXD | via G14 → G31 | insulated core · cut 70 mm, strip 3 mm both ends |
| 8.5 | Link | J32 **H2** SIM TXD → SIM868 pin 2 | K13 **ESP32-S3** J1-11 GPIO18<br>H2 SIM TXD → GPIO18 (ESP RX) | via E32 → E13 | insulated core · cut 85 mm, strip 3 mm both ends |
| 8.6 | 0.5 mm² | L35 VBAT rail, next to C1+ | ↗ SIM868 battery socket + (PH2.0 plug, red lead)<br>SIM868 power: straight from the switched 18650, never 5 V | | 0.5 mm² red · cut 80 mm, strip 4 mm, tin |
| 8.7 | 0.5 mm² | L33 GND bus col 33, next to C1– | ↗ SIM868 battery socket – (PH2.0 plug, black lead) | | 0.5 mm² black · cut 75 mm, strip 4 mm, tin |
| 8.8 | Jumper | J29 H2 3V3 | ↗ SIM868 header pin 36 (3V3(OUT))<br>female end on H2, male end straight into the SIM868 socket (Pico removed) | | M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket |
| 8.9 | Jumper | J30 H2 PWRKEY | ↗ SIM868 header pin 19 (GP14 / PWR_EN)<br>female end on H2, male end straight into the SIM868 socket (Pico removed) | | M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket |
| 8.10 | Jumper | J31 H2 SIM RXD | ↗ SIM868 header pin 1 (GP0 / RXD)<br>female end on H2, male end straight into the SIM868 socket (Pico removed) | | M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket |

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 8.11 | Jumper | J32 H2 SIM TXD | ↗ SIM868 header pin 2 (GP1 / TXD)<br>female end on H2, male end straight into the SIM868 socket (Pico removed) | | M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket |
| 8.12 | Jumper | J33 H2 GND | ↗ SIM868 header pin 3 (GND)<br>female end on H2, male end straight into the SIM868 socket (Pico removed) | | M-F jumper 10 cm (stock row 39), used as is: female end on H2, male end into the SIM868 socket |




SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT




COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

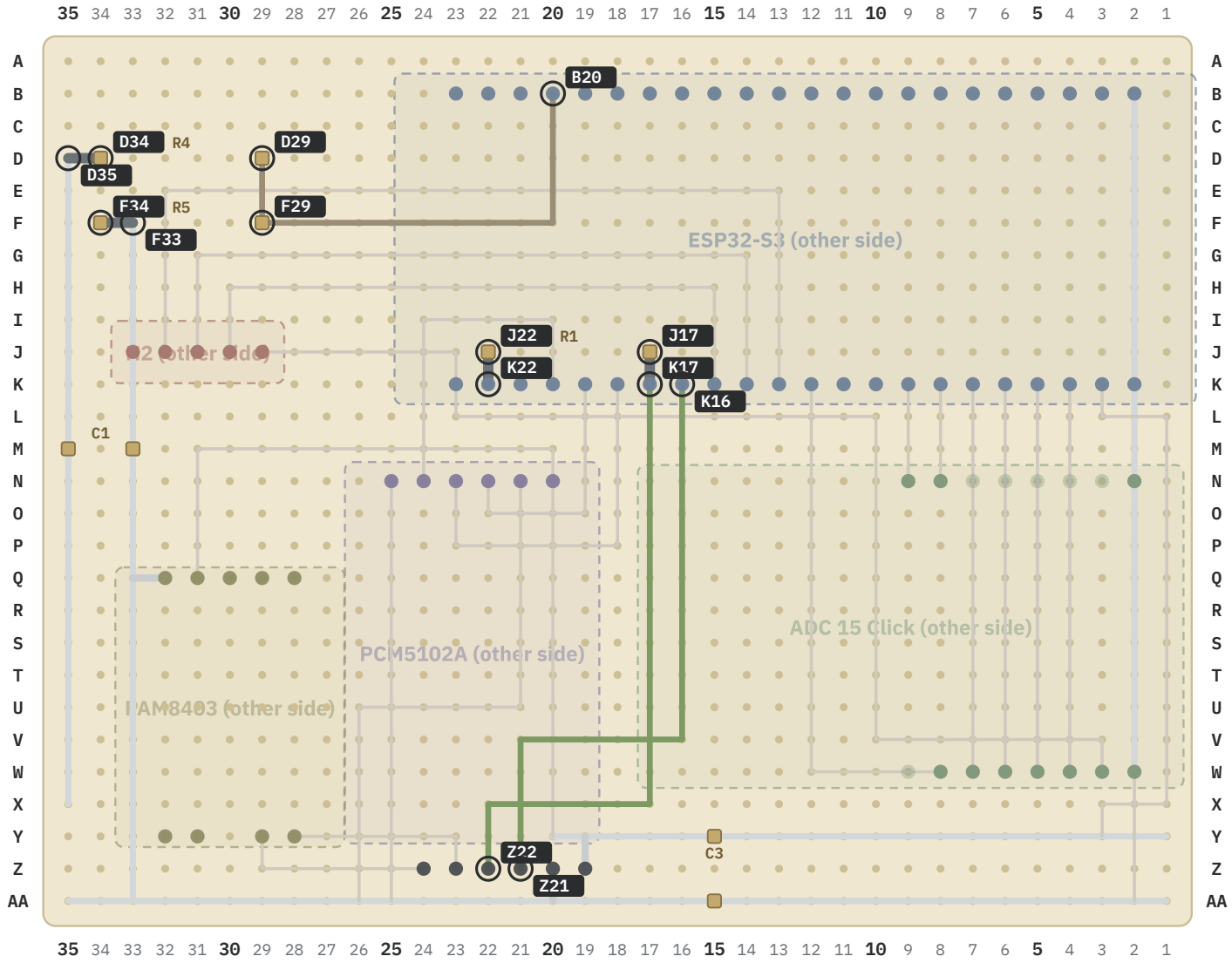
## STEP 9 Battery sense, button & LED lines

Soldering station

R4, R5 (free area at the back right, rows D and F) and R1 (under the ESP32-S3) go in from the component side; bridges and links on the solder side.

DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 9.1 | Part | **R1** 10 kΩ · button pull-up<br>legs into J17 J22 (component side) | | | bend legs flat after soldering |
| 9.2 | Part | **R4** 100 kΩ · battery divider, top<br>legs into D34 D29 (component side) | | | bend legs flat after soldering |
| 9.3 | Part | **R5** 100 kΩ · battery divider, bottom<br>legs into F29 F34 (component side) | | | bend legs flat after soldering |
| 9.4 | Bridge | J17 **R1** 10 kΩ leg | K17 **ESP32-S3** J1-7 GPIO7<br>R1 leg → GPIO7 | | solder bridge / bent leg |
| 9.5 | Bridge | J22 **R1** 10 kΩ leg | K22 **ESP32-S3** J1-2 3V3<br>R1 leg → 3V3 (J1-2) | | solder bridge / bent leg |
| 9.6 | Bridge | D34 **R4** 100 kΩ leg | D35 **VBAT rail**<br>R4 leg → VBAT rail | | solder bridge / bent leg |
| 9.7 | Bridge | F34 **R5** 100 kΩ leg | F33 **GND bus** col 33<br>R5 leg → GND bus col 33 | | solder bridge / bent leg |
| 9.8 | Link | K17 **ESP32-S3** J1-7 GPIO7 | Z22 **H1** BTN<br>button line | via X17 → X22 | insulated core · cut 60 mm, strip 3 mm both ends |
| 9.9 | Link | K16 **ESP32-S3** J1-8 GPIO15 | Z21 **H1** LED<br>LED data | via V16 → V21 | insulated core · cut 60 mm, strip 3 mm both ends |
| 9.10 | Link | D29 **R4** 100 kΩ leg | F29 **R5** 100 kΩ leg<br>divider node R4–R5 | straight | insulated core · cut 15 mm, strip 3 mm both ends |
| 9.11 | Link | F29 **R5** 100 kΩ leg | B20 **ESP32-S3** J3-4 GPIO1<br>VBAT ÷ 2 → GPIO1 | via F20 | insulated core · cut 45 mm, strip 3 mm both ends |




SOLDER SIDE · board turned over like a page · row A at the top · column 1 on the RIGHT

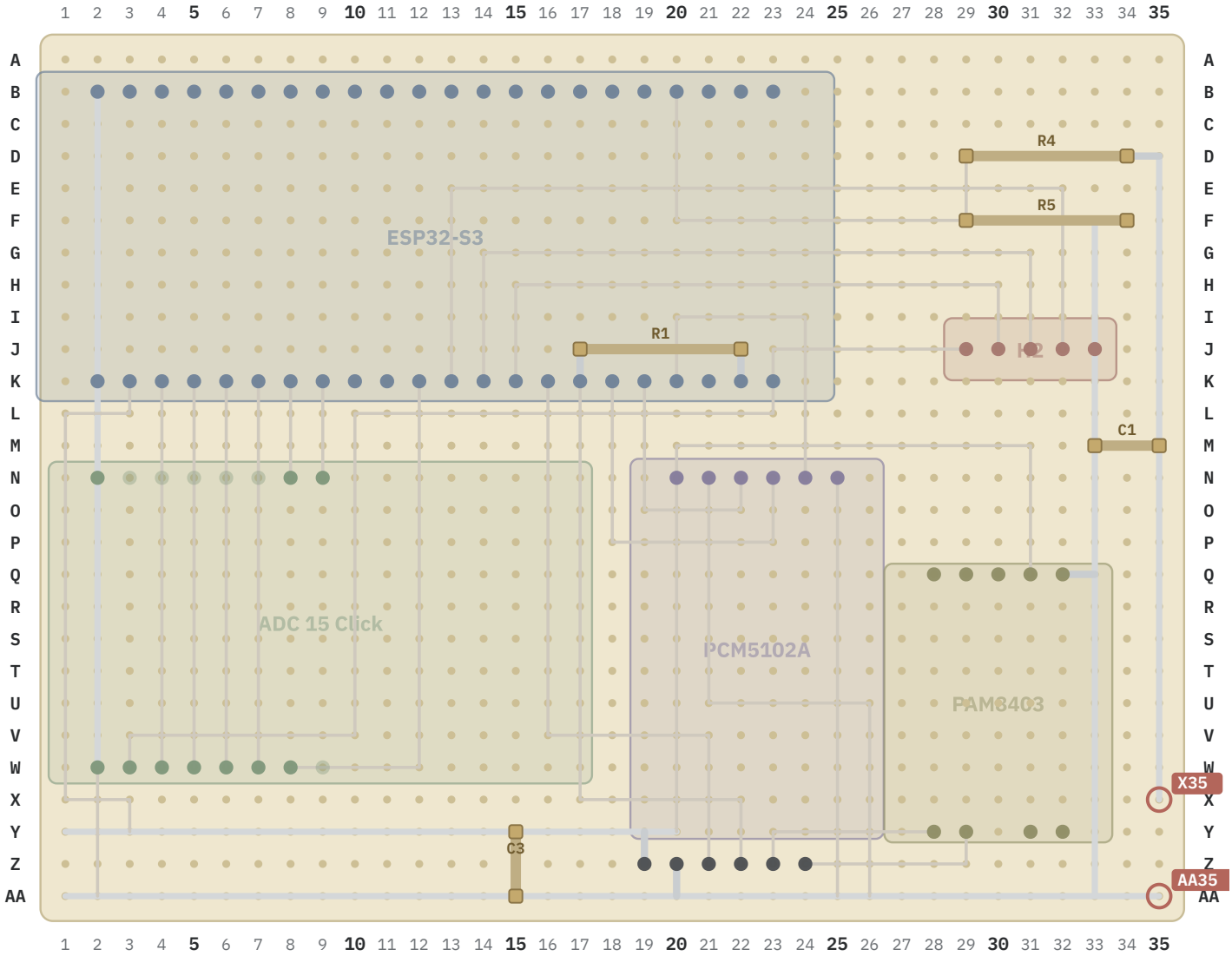
## STEP 10 Battery → switch

Soldering station

Battery out of the holder. Push the rocker into the base's wall cutout from outside first, then solder its two pins (lamp pin unused). Rail-end holes: tin the 0.5 mm² wire, push it through the hole from the component side and solder it to the rail underneath. No charger board in the device: charge the 18650 in an external charger and use a protected cell (nothing else stops over-discharge).

DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 10.1 | Lead | ↗ Holder + lead | ↗ Rocker pin 1 | | holder lead · trim to 65 mm, strip 4 mm |
| 10.2 | Lead | ↗ Holder – lead | AA35 GND rail end | | holder lead · trim to 65 mm, strip 4 mm |
| 10.3 | 0.5 mm² | ↗ Rocker pin 2 | X35 VBAT rail end | | 0.5 mm² red · cut 60 mm, strip 4 mm, tin |




COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

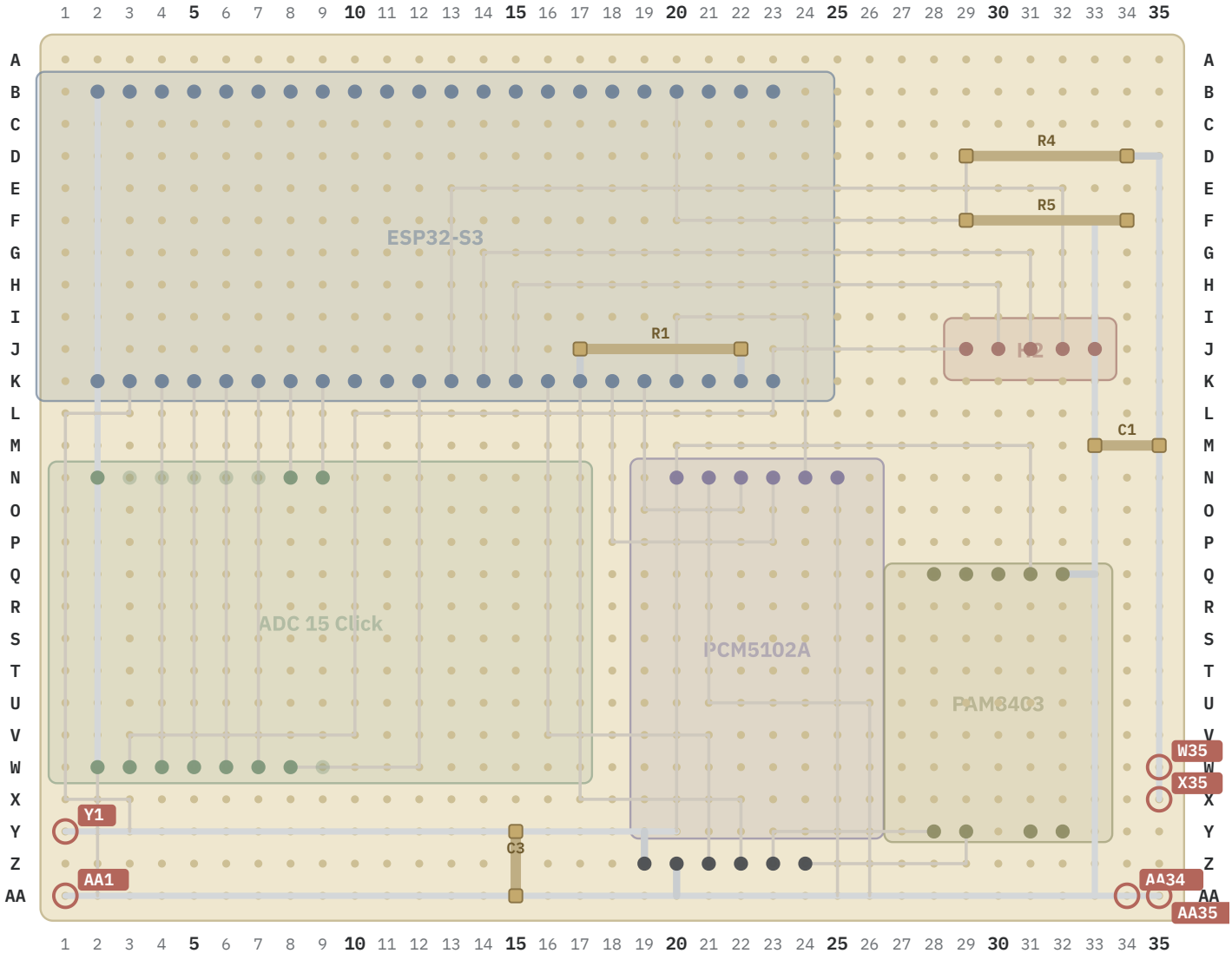
## STEP 11 Boost converter to 5.0 V

Multimeter · Soldering station

Before connecting the output: power the boost input from the battery, turn its trimmer until the output reads 5.00 V (±0.05 V). Then wire it.

DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 11.1 | 0.5 mm² | W35 VBAT rail | ↗ Boost IN+ | | 0.5 mm² red · cut 90 mm, strip 4 mm, tin |
| 11.2 | 0.5 mm² | AA34 GND rail | ↗ Boost IN− | | 0.5 mm² black · cut 95 mm, strip 4 mm, tin |
| 11.3 | 0.5 mm² | ↗ Boost OUT+ | Y1 5 V rail start | | 0.5 mm² red · cut 120 mm, strip 4 mm, tin |
| 11.4 | 0.5 mm² | ↗ Boost OUT− | AA1 GND rail start | | 0.5 mm² black · cut 125 mm, strip 4 mm, tin |




COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

# STEP 12 Lid board: button, LED, diode

Soldering station · Cutting bench

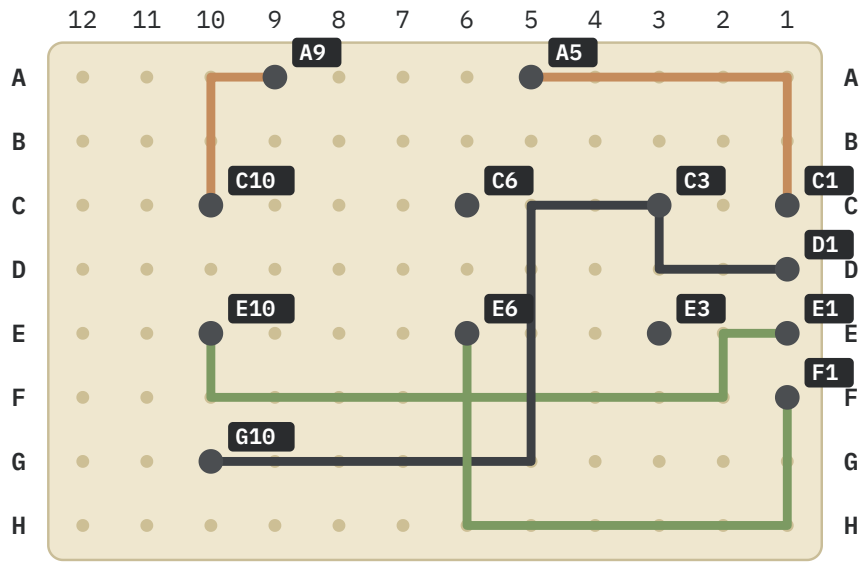
Lid board, component side up: fit the parts into the listed holes, then turn it over and make the links. The header pins are bent 90° outward over the edge after soldering.

## PARTS INTO HOLES (COMPONENT SIDE)

| | |
|---|---|
| C1 D1 E1 F1 | Lid header, 4 straight pins (C1 5V, D1 GND, E1 DIN, F1 BTN), bent 90° outward over the edge |
| C3 C6 E3 E6 | Tact switch legs (C3 + C6 are joined inside, E3 + E6 are joined inside) |
| A5 → A9 | 1N4001: anode A5, cathode (stripe) A9 |
| C10 E10 G10 | WS2812B single pixel: stubs (resistor-leg offcuts) from its +5V, DIN, GND pads into C10, E10, G10 |

## DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 12.1 | Link | C1 **Lid header** 5V | A5 **1N4001** anode<br>header 5V → diode anode | via A1 | insulated core · cut 25 mm, strip 3 mm both ends |
| 12.2 | Link | A9 **1N4001** cathode (stripe) | C10 **WS2812B** +5V stub<br>diode cathode → pixel +5V | via A10 | insulated core · cut 20 mm, strip 3 mm both ends |
| 12.3 | Link | E1 **Lid header** DIN | E10 **WS2812B** DIN stub<br>header DIN → pixel DIN | via E2 → F2 → F10 | insulated core · cut 40 mm, strip 3 mm both ends |
| 12.4 | Link | D1 **Lid header** GND | C3 **Button** leg 1<br>header GND → button leg 1 | via D3 | insulated core · cut 20 mm, strip 3 mm both ends |
| 12.5 | Link | C3 **Button** leg 1 | G10 **WS2812B** GND stub<br>button leg 1 → pixel GND | via C5 → G5 | insulated core · cut 40 mm, strip 3 mm both ends |
| 12.6 | Link | F1 **Lid header** BTN | E6 **Button** leg 4<br>header BTN → button leg 4 (diagonal) | via H1 → H6 | insulated core · cut 35 mm, strip 3 mm both ends |




LID BOARD · LINK SIDE (turned over like a page: column 1 on the RIGHT)

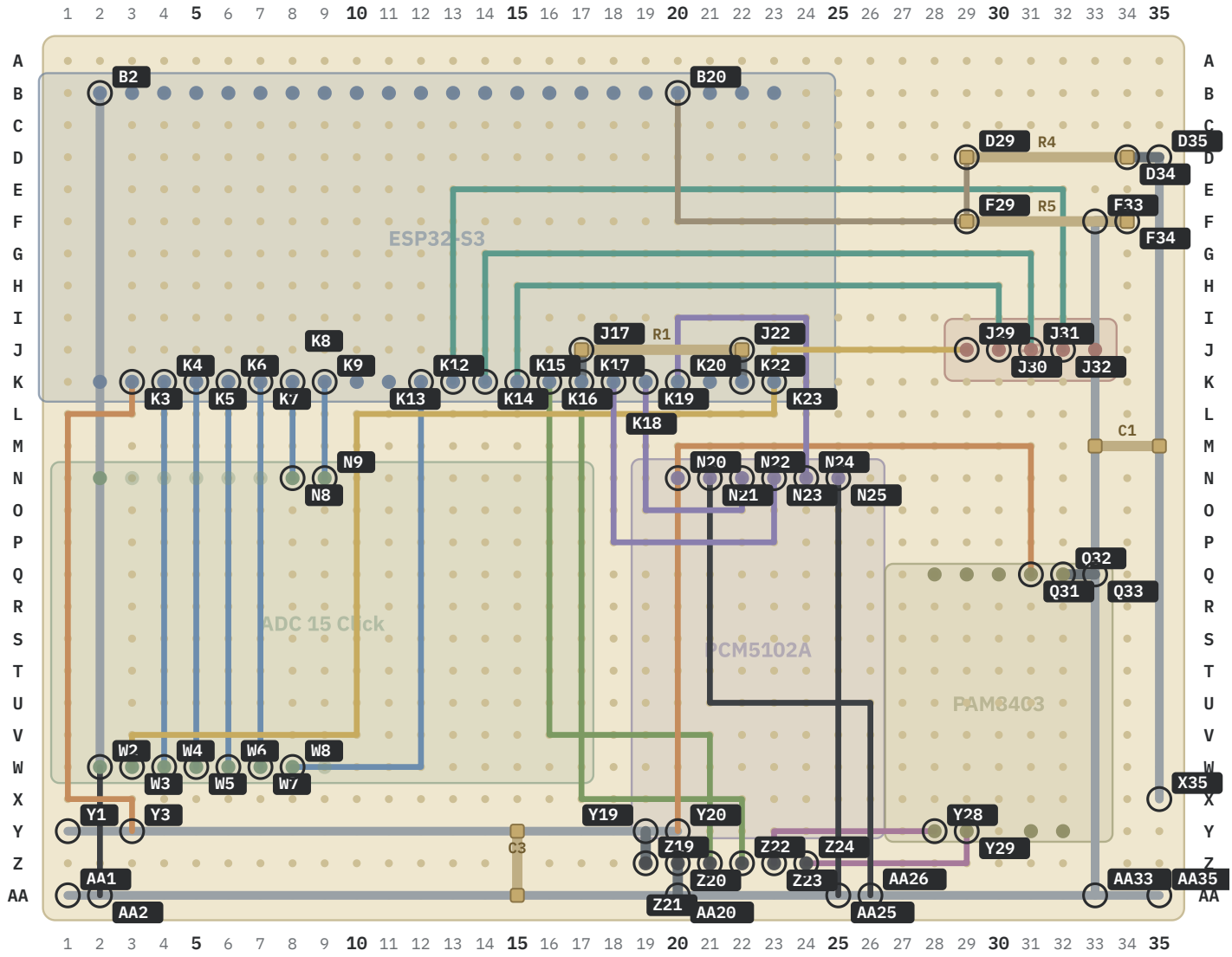
## STEP 13 Lid harness & speaker leads

Cutting bench · Soldering station

Four F-F jumpers are shortened (cut the stated length out of the middle, strip 4 mm, twist, solder, heat-shrink) and two are cut for the speaker (keep one female plug, solder the cut end to the speaker tab, heat-shrink).

DO THESE IN ORDER

| # | TYPE | FROM | TO | ROUTE (SOLDER SIDE) | WIRE / CUT |
|---|---|---|---|---|---|
| 13.1 | Jumper | H1 5V Z19 | ↗ Lid header 5V | | F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink |
| 13.2 | Jumper | H1 GND Z20 | ↗ Lid header GND | | F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink |
| 13.3 | Jumper | H1 LED Z21 | ↗ Lid header DIN | | F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink |
| 13.4 | Jumper | H1 BTN Z22 | ↗ Lid header BTN | | F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink |
| 13.5 | Jumper | H1 SPK+ Z23 | ↗ Speaker + tab | | F-F jumper → 75 mm: keep one female end, cut 25 mm off, solder to the speaker tab |
| 13.6 | Jumper | H1 SPK– Z24 | ↗ Speaker – tab | | F-F jumper → 90 mm: keep one female end, cut 10 mm off, solder to the speaker tab |




COMPONENT SIDE · row A (back) at the top · column 1 on the left · solder-side wires shown through the board

## STEP 14 Check sheet

Battery out, all modules unplugged. Beep each pair with the meter. Tick when done.

### MUST NOT BEEP (SHORTS)

| | | |
|---|---|---|
| ☐ | AA1 ↔ Y1 | GND ↔ 5 V |
| ☐ | AA1 ↔ X35 | GND ↔ VBAT |
| ☐ | AA1 ↔ K23 | GND ↔ 3V3 |
| ☐ | Y1 ↔ X35 | 5 V ↔ VBAT |
| ☐ | Y1 ↔ L35 | 5 V ↔ SIM868 VBAT lead (would destroy the SIM868) |
| ☐ | Y1 ↔ J29 | 5 V ↔ H2 3V3 |
| ☐ | L35 ↔ J29 | VBAT ↔ H2 3V3 |
| ☐ | Y28 ↔ AA1 | PAM L+ ↔ GND |
| ☐ | Y29 ↔ AA1 | PAM L− ↔ GND |

### MUST BEEP (EVERY LINK, END TO END)

| | | | |
|---|---|---|---|
| ☐ | Z19 ↔ Y19 | H1 5V → 5 V rail | step 4 |
| ☐ | Z20 ↔ AA20 | H1 GND → GND rail | step 4 |
| ☐ | Q32 ↔ Q33 | PAM8403 GND → GND bus | step 4 |
| ☐ | J17 ↔ K17 | R1 leg → GPIO7 | step 9 |
| ☐ | J22 ↔ K22 | R1 leg → 3V3 (J1-2) | step 9 |
| ☐ | D34 ↔ D35 | R4 leg → VBAT rail | step 9 |
| ☐ | F34 ↔ F33 | R5 leg → GND bus col 33 | step 9 |
| ☐ | K6 ↔ W6 | SCK | step 5 |
| ☐ | K5 ↔ W5 | MISO | step 5 |
| ☐ | K4 ↔ W4 | MOSI | step 5 |
| ☐ | K7 ↔ W7 | ADC chip-select | step 5 |
| ☐ | K9 ↔ N9 | clock-chip select | step 5 |
| ☐ | K8 ↔ N8 | DRDY, active low | step 5 |
| ☐ | K12 ↔ W8 | ADC reset | step 5 |
| ☐ | K17 ↔ Z22 | button line | step 9 |
| ☐ | K16 ↔ Z21 | LED data | step 9 |
| ☐ | Y28 ↔ Z23 | speaker + | step 7 |
| ☐ | Y29 ↔ Z24 | speaker −, NOT ground | step 7 |
| ☐ | W2 ↔ AA2 | GND bus col 2 → GND rail | step 4 |
| ☐ | Y3 ↔ K3 | 5 V rail → ESP32 5V | step 4 |
| ☐ | Y20 ↔ N20 | 5 V rail → PCM VIN | step 4 |
| ☐ | N20 ↔ Q31 | PCM VIN → PAM8403 +5V | step 4 |
| ☐ | N21 ↔ AA26 | PCM GND → GND rail | step 4 |
| ☐ | K23 ↔ W3 | ESP32 3V3 → Click 3V3 | step 4 |
| ☐ | K20 ↔ N24 | bit clock | step 7 |
| ☐ | K19 ↔ N22 | word clock | step 7 |
| ☐ | K18 ↔ N23 | data | step 7 |
| ☐ | N25 ↔ AA25 | SCK → GND rail (SCK must be grounded) | step 7 |
| ☐ | K23 ↔ J29 | ESP32 3V3 (J1-1) → H2 3V3 (feeds the SIM868 level shifter) | step 8 |
| ☐ | K15 ↔ J30 | GPIO16 → H2 PWRKEY (HIGH 1.5 s = power on/off) | step 8 |
| ☐ | K14 ↔ J31 | GPIO17 (ESP TX) → H2 SIM RXD | step 8 |
| ☐ | J32 ↔ K13 | H2 SIM TXD → GPIO18 (ESP RX) | step 8 |
| ☐ | D29 ↔ F29 | divider node R4–R5 | step 9 |

| | | | |
|---|---|---|---|
| ☐ | `F29` ↔ `B20` | VBAT ÷ 2 → GPIO1 | step 9 |
| ☐ | `C1` ↔ `A5` (lid) | header 5V → diode anode | step 12 |
| ☐ | `A9` ↔ `C10` (lid) | diode cathode → pixel +5V | step 12 |
| ☐ | `E1` ↔ `E10` (lid) | header DIN → pixel DIN | step 12 |
| ☐ | `D1` ↔ `C3` (lid) | header GND → button leg 1 | step 12 |
| ☐ | `C3` ↔ `G10` (lid) | button leg 1 → pixel GND | step 12 |
| ☐ | `F1` ↔ `E6` (lid) | header BTN → button leg 4 (diagonal) | step 12 |

### THEN, MODULES PLUGGED IN, BATTERY IN, ROCKER ON

- ☐ 5.0 V between Z19 and Z20 (H1 5V–GND)
- ☐ 3.3 V between W3 and W2 (Click 3V3–GND)
- ☐ 3.3 V between J29 and J33 (H2 3V3–GND)
- ☐ 3.7–4.2 V between L35 and L33 (SIM868 battery lead) and the same on the PH2.0 plug, red = +
- ☐ SIM868: PWRKEY pulse → NET LED blinks fast, then about every 3 s (registered); AT → AT+CPIN? → AT+CREG? (0,1) → test call
- ☐ Rocker OFF before plugging USB-C into the ESP32-S3 to flash

## CUT Cut list (every wire cut to length)

| STEP | TYPE | FROM | TO | LENGTH | INSTRUCTION |
|---|---|---|---|---|---|
| 4 | Bare wire | AA1 **GND rail** | AA35 **GND rail** | **100 mm** | bare 20 AWG · cut 100 mm |
| 4 | Bare wire | Y1 **5 V rail** | Y20 **5 V rail** | **60 mm** | bare 20 AWG · cut 60 mm |
| 4 | Bare wire | B2 **ESP32-S3** J3-22 GND | W2 **Click** GND | **65 mm** | bare 20 AWG · cut 65 mm |
| 4 | Bare wire | F33 **GND bus** col 33 | AA33 **GND rail** | **65 mm** | bare 20 AWG · cut 65 mm |
| 4 | Bare wire | D35 **VBAT rail** | X35 **VBAT rail** | **65 mm** | bare 20 AWG · cut 65 mm |
| 4 | Link | W2 **Click** GND | AA2 **GND rail** | **20 mm** | insulated core · cut 20 mm, strip 3 mm both ends |
| 4 | Link | Y3 **5 V rail** | K3 **ESP32-S3** J1-21 5V | **55 mm** | insulated core · cut 55 mm, strip 3 mm both ends |
| 4 | Link | Y20 **5 V rail** | N20 **PCM5102A** VIN | **40 mm** | insulated core · cut 40 mm, strip 3 mm both ends |
| 4 | Link | N20 **PCM5102A** VIN | Q31 **PAM8403** +5V | **50 mm** | insulated core · cut 50 mm, strip 3 mm both ends |
| 4 | Link | N21 **PCM5102A** GND | AA26 **GND rail** | **55 mm** | insulated core · cut 55 mm, strip 3 mm both ends |
| 4 | Link | K23 **ESP32-S3** J1-1 3V3 | W3 **Click** 3V3 | **90 mm** | insulated core · cut 90 mm, strip 3 mm both ends |
| 5 | Link | K6 **ESP32-S3** J1-18 GPIO12 | W6 **Click** SCK | **40 mm** | insulated core · cut 40 mm, strip 3 mm both ends |
| 5 | Link | K5 **ESP32-S3** J1-19 GPIO13 | W5 **Click** SDO (MISO) | **40 mm** | insulated core · cut 40 mm, strip 3 mm both ends |
| 5 | Link | K4 **ESP32-S3** J1-20 GPIO14 | W4 **Click** SDI (MOSI) | **40 mm** | insulated core · cut 40 mm, strip 3 mm both ends |
| 5 | Link | K7 **ESP32-S3** J1-17 GPIO11 | W7 **Click** CS1 | **40 mm** | insulated core · cut 40 mm, strip 3 mm both ends |
| 5 | Link | K9 **ESP32-S3** J1-15 GPIO9 | N9 **Click** CS2 | **20 mm** | insulated core · cut 20 mm, strip 3 mm both ends |
| 5 | Link | K8 **ESP32-S3** J1-16 GPIO10 | N8 **Click** INT (DRDY) | **20 mm** | insulated core · cut 20 mm, strip 3 mm both ends |
| 5 | Link | K12 **ESP32-S3** J1-12 GPIO8 | W8 **Click** RST | **50 mm** | insulated core · cut 50 mm, strip 3 mm both ends |
| 6 | Pair | ↗ SM-24 + tab | ↗ Click CH1+ screw terminal | **115 mm** | 6×0.22 cable core · cut 115 mm, twist the two cores ~1 turn/cm |
| 6 | Pair | ↗ SM-24 – tab | ↗ Click CH1– screw terminal | **105 mm** | 6×0.22 cable core · cut 105 mm, twist the two cores ~1 turn/cm |
| 6 | Top wire | AA18 GND rail | ↗ Click CH1– screw terminal | **55 mm** | insulated core · cut 55 mm, strip 5 mm (screw end) / 3 mm |
| 6 | Top wire | AA19 GND rail | ↗ Click CH2– screw terminal | **40 mm** | insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm |
| 6 | Top wire | ↗ Click CH2+ screw terminal | ↗ Click CH2– screw terminal | **20 mm** | insulated core · cut 20 mm, strip 5 mm (screw end) / 3 mm |
| 7 | Link | Y28 **PAM8403** L+ out | Z23 **H1** SPK+ | **25 mm** | insulated core · cut 25 mm, strip 3 mm both ends |
| 7 | Link | Y29 **PAM8403** L– out | Z24 **H1** SPK– | **25 mm** | insulated core · cut 25 mm, strip 3 mm both ends |
| 7 | Link | K20 **ESP32-S3** J1-4 GPIO4 | N24 **PCM5102A** BCK | **40 mm** | insulated core · cut 40 mm, strip 3 mm both ends |
| 7 | Link | K19 **ESP32-S3** J1-5 GPIO5 | N22 **PCM5102A** LCK | **30 mm** | insulated core · cut 30 mm, strip 3 mm both ends |

| STEP | TYPE | FROM | TO | LENGTH | INSTRUCTION |
|---|---|---|---|---|---|
| 7 | Link | K18 **ESP32-S3** J1-6 GPIO6 | N23 **PCM5102A** DIN | **40 mm** | insulated core · cut 40 mm, strip 3 mm both ends |
| 7 | Link | N25 **PCM5102A** SCK | AA25 **GND rail** | **45 mm** | insulated core · cut 45 mm, strip 3 mm both ends |
| 7 | Top wire | ↗ PCM5102A pad L (top) | ↗ PAM8403 pad L in (top, above Q28) | **40 mm** | insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm |
| 7 | Top wire | ↗ PCM5102A pad AGND (top) | ↗ PAM8403 pad ⏚ in (top, above Q29) | **40 mm** | insulated core · cut 40 mm, strip 5 mm (screw end) / 3 mm |
| 8 | Link | K23 **ESP32-S3** J1-1 3V3 | J29 **H2** 3V3 → SIM868 pin 36 | **30 mm** | insulated core · cut 30 mm, strip 3 mm both ends |
| 8 | Link | K15 **ESP32-S3** J1-9 GPIO16 | J30 **H2** PWRKEY → SIM868 pin 19 | **60 mm** | insulated core · cut 60 mm, strip 3 mm both ends |
| 8 | Link | K14 **ESP32-S3** J1-10 GPIO17 | J31 **H2** SIM RXD → SIM868 pin 1 | **70 mm** | insulated core · cut 70 mm, strip 3 mm both ends |
| 8 | Link | J32 **H2** SIM TXD → SIM868 pin 2 | K13 **ESP32-S3** J1-11 GPIO18 | **85 mm** | insulated core · cut 85 mm, strip 3 mm both ends |
| 8 | 0.5 mm² | L35 VBAT rail, next to C1+ | ↗ SIM868 battery socket + (PH2.0 plug, red lead) | **5 mm** | 0.5 mm² red · cut 80 mm, strip 4 mm, tin |
| 8 | 0.5 mm² | L33 GND bus col 33, next to C1− | ↗ SIM868 battery socket − (PH2.0 plug, black lead) | **5 mm** | 0.5 mm² black · cut 75 mm, strip 4 mm, tin |
| 9 | Link | K17 **ESP32-S3** J1-7 GPIO7 | Z22 **H1** BTN | **60 mm** | insulated core · cut 60 mm, strip 3 mm both ends |
| 9 | Link | K16 **ESP32-S3** J1-8 GPIO15 | Z21 **H1** LED | **60 mm** | insulated core · cut 60 mm, strip 3 mm both ends |
| 9 | Link | D29 **R4** 100 kΩ leg | F29 **R5** 100 kΩ leg | **15 mm** | insulated core · cut 15 mm, strip 3 mm both ends |
| 9 | Link | F29 **R5** 100 kΩ leg | B20 **ESP32-S3** J3-4 GPIO1 | **45 mm** | insulated core · cut 45 mm, strip 3 mm both ends |
| 10 | Lead | ↗ Holder + lead | ↗ Rocker pin 1 | **65 mm** | holder lead · trim to 65 mm, strip 4 mm |
| 10 | Lead | ↗ Holder − lead | AA35 GND rail end | **65 mm** | holder lead · trim to 65 mm, strip 4 mm |
| 10 | 0.5 mm² | ↗ Rocker pin 2 | X35 VBAT rail end | **5 mm** | 0.5 mm² red · cut 60 mm, strip 4 mm, tin |
| 11 | 0.5 mm² | W35 VBAT rail | ↗ Boost IN+ | **5 mm** | 0.5 mm² red · cut 90 mm, strip 4 mm, tin |
| 11 | 0.5 mm² | AA34 GND rail | ↗ Boost IN− | **5 mm** | 0.5 mm² black · cut 95 mm, strip 4 mm, tin |
| 11 | 0.5 mm² | ↗ Boost OUT+ | Y1 5 V rail start | **5 mm** | 0.5 mm² red · cut 120 mm, strip 4 mm, tin |
| 11 | 0.5 mm² | ↗ Boost OUT− | AA1 GND rail start | **5 mm** | 0.5 mm² black · cut 125 mm, strip 4 mm, tin |
| 12 | Link | C1 **Lid header** 5V | A5 **1N4001** anode | **25 mm** | insulated core · cut 25 mm, strip 3 mm both ends |
| 12 | Link | A9 **1N4001** cathode (stripe) | C10 **WS2812B** +5V stub | **20 mm** | insulated core · cut 20 mm, strip 3 mm both ends |
| 12 | Link | E1 **Lid header** DIN | E10 **WS2812B** DIN stub | **40 mm** | insulated core · cut 40 mm, strip 3 mm both ends |
| 12 | Link | D1 **Lid header** GND | C3 **Button** leg 1 | **20 mm** | insulated core · cut 20 mm, strip 3 mm both ends |
| 12 | Link | C3 **Button** leg 1 | G10 **WS2812B** GND stub | **40 mm** | insulated core · cut 40 mm, strip 3 mm both ends |
| 12 | Link | F1 **Lid header** BTN | E6 **Button** leg 4 | **35 mm** | insulated core · cut 35 mm, strip 3 mm both ends |

| STEP | TYPE | FROM | TO | LENGTH | INSTRUCTION |
|---|---|---|---|---|---|
| 13 | Jumper | H1 5V Z19 | ↗ Lid header 5V | **60 mm** | F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink |
| 13 | Jumper | H1 GND Z20 | ↗ Lid header GND | **60 mm** | F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink |
| 13 | Jumper | H1 LED Z21 | ↗ Lid header DIN | **60 mm** | F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink |
| 13 | Jumper | H1 BTN Z22 | ↗ Lid header BTN | **60 mm** | F-F jumper → 60 mm: cut 40 mm out of the middle, splice + heat-shrink |
| 13 | Jumper | H1 SPK+ Z23 | ↗ Speaker + tab | **75 mm** | F-F jumper → 75 mm: keep one female end, cut 25 mm off, solder to the speaker tab |
| 13 | Jumper | H1 SPK– Z24 | ↗ Speaker – tab | **90 mm** | F-F jumper → 90 mm: keep one female end, cut 10 mm off, solder to the speaker tab |

Totals: insulated cores ≈ 1370 mm (≈ 280 mm of the 6-core cable), bare 20 AWG ≈ 355 mm. Module outlines on the maps are approximate; hole positions are exact.